

TASK -1

IPv4 vs IPv6 Addressing Comparison

IPv4	IPv6
Uses 32-bit addresses.	Uses 128-bit addresses.
Written in decimal format (e.g., 192.168.1.1).	Written in hexadecimal format (e.g., 2001:0db8:85a3::8a2e:0370:7334).
Supports about 4.3 billion unique addresses.	Supports approximately 340 undecillion (3.4×10^{38}) unique addresses.
Header size is 20 bytes and more complex.	Has a 40-byte fixed header that is simpler and more efficient.
Uses NAT (Network Address Translation) because of limited addresses.	Generally does not require NAT due to the large address space.
Broadcast communication is supported.	No broadcast ; uses multicast and anycast instead.
Address configuration can be manual or via DHCP .	Supports SLAAC (Stateless Address Autoconfiguration) and DHCPv6.
Security (IPsec) is optional .	IPsec support is built into the protocol (though use depends on implementation).

TASK-2

Static IPv4 Address Assignment (Network: 192.168.10.0/24)

Network Address: 192.168.10.0/24

Subnet Mask: 255.255.255.0

Default Gateway (Router): 192.168.10.1

Device	IP Address	Subnet Mask	Default Gateway
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Phone	192.168.10.11	255.255.255.0	192.168.10.1
Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
Printer	192.168.10.13	255.255.255.0	192.168.10.1

Notes

- All devices are assigned **unique static IP addresses** within the **192.168.10.0/24** network.
- The **subnet mask (255.255.255.0)** indicates that all devices belong to the same local network.
- The **default gateway (192.168.10.1)** is the home Wi-Fi router, which allows the devices to access the internet and communicate with other networks.

TASK-3

Subnet Calculation for 192.168.1.0/27

- **Network:** 192.168.1.0
- **CIDR Notation:** /27
- **Subnet Mask:** 255.255.255.224

Step 1: Number of Valid Subnets

A /27 mask means **3 bits** are borrowed from the host portion (default /24).

- **Number of subnets = $2^3 = 8$ subnets**

Step 2: Hosts Per Subnet

A /27 mask leaves **5 host bits**.

- **Total addresses per subnet = $2^5 = 32$**
- **Usable host addresses = $32 - 2 = 30$ hosts**

Summary

Item	Value
Network Address	192.168.1.0
CIDR	/27
Subnet Mask	255.255.255.224
Number of Subnets	8
Addresses per Subnet	32
Usable Hosts per Subnet	30

First Three Valid Subnets

Subnet	Network Address	Usable Host Range	Broadcast Address
Subnet 1	192.168.1.0/27	192.168.1.1 – 192.168.1.30	192.168.1.31
Subnet 2	192.168.1.32/27	192.168.1.33 – 192.168.1.62	192.168.1.63
Subnet 3	192.168.1.64/27	192.168.1.65 – 192.168.1.94	192.168.1.95

All /27 Subnets (for reference)

Subnet	Network	Broadcast
1	192.168.1.0	192.168.1.31
2	192.168.1.32	192.168.1.63
3	192.168.1.64	192.168.1.95
4	192.168.1.96	192.168.1.127
5	192.168.1.128	192.168.1.159
6	192.168.1.160	192.168.1.191
7	192.168.1.192	192.168.1.223
8	192.168.1.224	192.168.1.255

TASK-4

Here is the step-by-step breakdown for the IP address **10.0.5.17** with the subnet mask **255.255.255.0**:

Step 1: Convert Subnet Mask to CIDR Notation

- **Subnet Mask:** 255.255.255.0
- **Binary Representation:** 11111111.11111111.11111111.00000000
- Counting the active network bits (the 1s) gives **24 bits**.
- **CIDR Notation:** /24

Step 2: Calculate the Network Address

Perform a bitwise **AND** operation between the IP address and the subnet mask:

- **IP Address (10.0.5.17):** 00001010.00000000.00000101.00010001
- **Subnet Mask (/24):** 11111111.11111111.11111111.00000000

- **Bitwise AND Result:** 00001010.00000000.00000101.00000000

Network Address = **10.0.5.0**

Step 3: Calculate the Broadcast Address

For a /24 network, set all **8 host bits** (the last octet) to **1**:

- **Network Portion (24 bits):** 10.0.5.
- **Host Portion (8 bits set to 1):** 11111111 (= 255)

Broadcast Address = **10.0.5.255**

Step 4: Determine the Valid Host Range

The valid host range falls between the Network Address and the Broadcast Address:

- **First Valid Host:** Network Address + 1 → 10.0.5.1
- **Last Valid Host:** Broadcast Address - 1 → 10.0.5.254

Summary Table

Item	Value
IP Address	10.0.5.17
Subnet Mask	255.255.255.0
CIDR Notation	/24
Network Address	10.0.5.0
Broadcast Address	10.0.5.255
Valid Host Range	10.0.5.1 to 10.0.5.254
Total Usable Hosts	254

IPv4 Subnet Calculator

Result

save

IP Address:	10.0.5.17
Network Address:	10.0.5.0
Usable Host IP Range:	10.0.5.1 - 10.0.5.254
Broadcast Address:	10.0.5.255
Total Number of Hosts:	256
Number of Usable Hosts:	254
Subnet Mask:	255.255.255.0
Wildcard Mask:	0.0.0.255
Binary Subnet Mask:	11111111.11111111.11111111.00000000
IP Class:	C
CIDR Notation:	/24
IP Type:	Private
Short:	10.0.5.17 /24
Binary ID:	000010100000000000000010100010001
Integer ID:	167773457
Hex ID:	0xa000511
in-addr.arpa:	17.5.0.10.in-addr.arpa
IPv4 Mapped Address:	::ffff:0a00.0511
6to4 Prefix:	2002:0a00.0511::/48

Network Class ☒ Any ☐ A ☐ B ☐ C

Subnet

255.255.255.0 /24

IP Address

10.0.5.17

Calculate

Clear